

Python is all you need: an overview of the composable, Python-native data stack

Deepyaman Datta · SciPy 2025 · July 9, 2025

Why should *you* care?

THE DATA SCIENCE **HIERARCHY OF NEEDS**

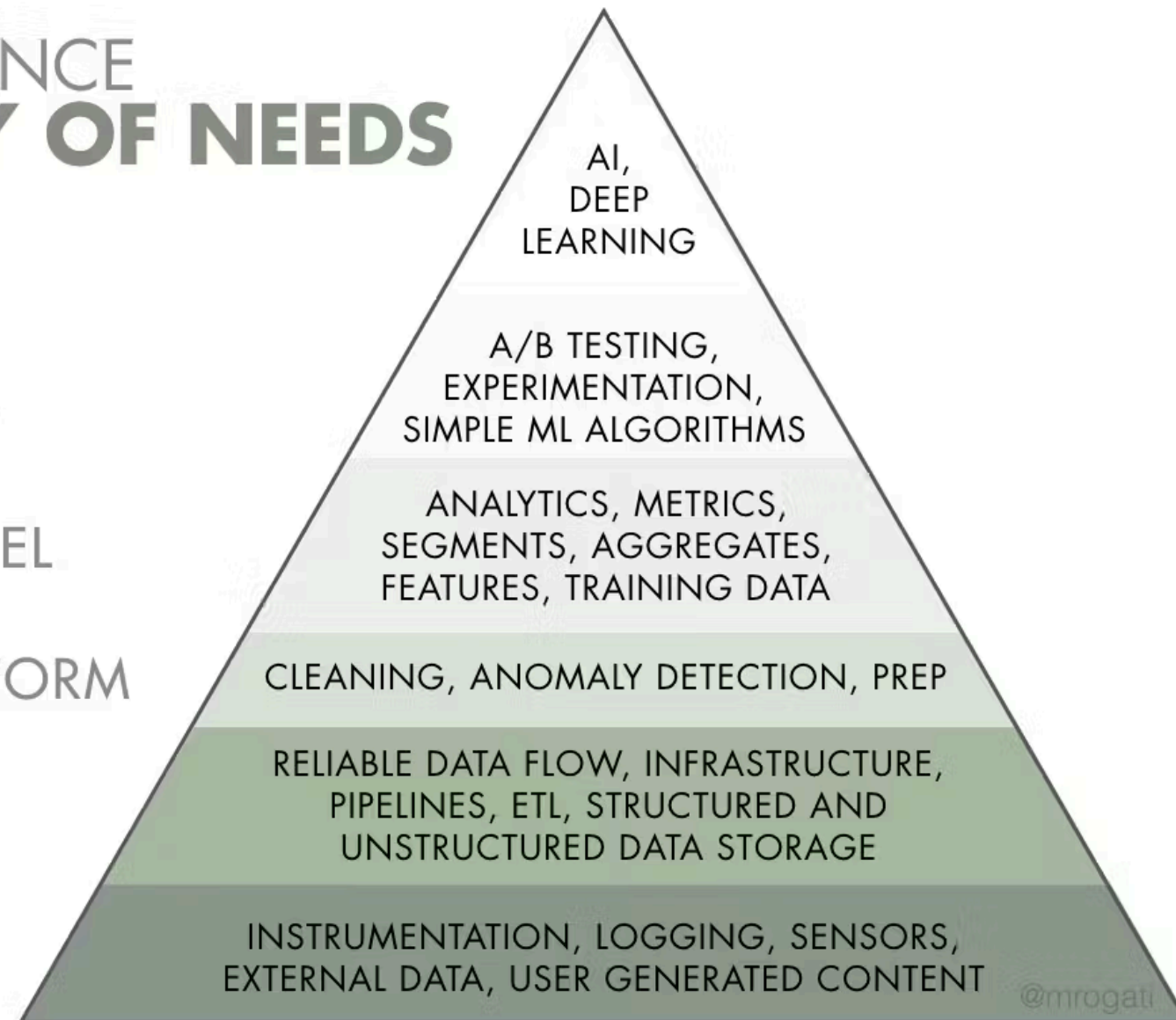
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

MOVE/STORE

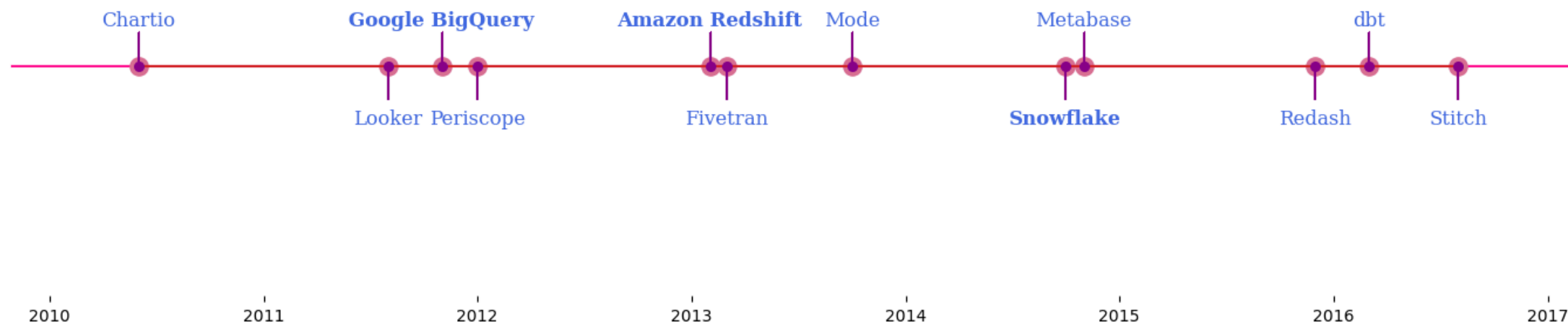
COLLECT



Source: The AI Hierarchy of Needs (Rogati, 2017)

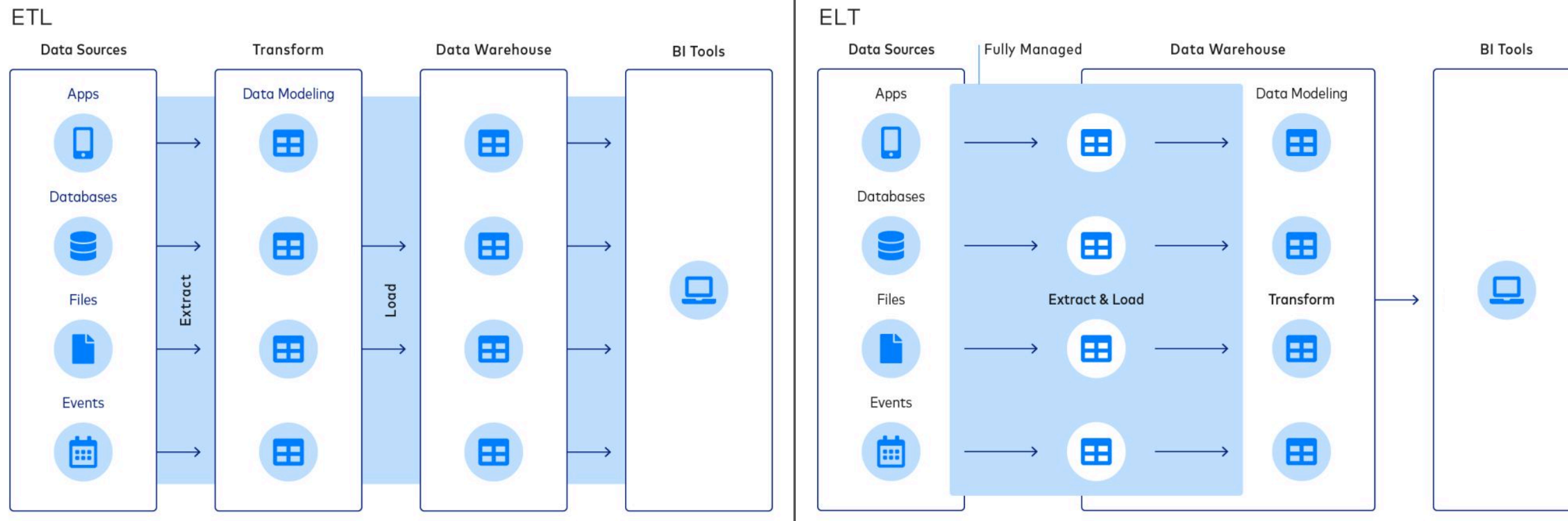
What is the modern data stack?

a set of products that a) redesigned the analytics workflow to take advantage of the cloud and b) all interacted with one another via SQL



ELT vs ETL

- ETL: transform before loading into the warehouse
- ELT: load first, then transform inside the warehouse

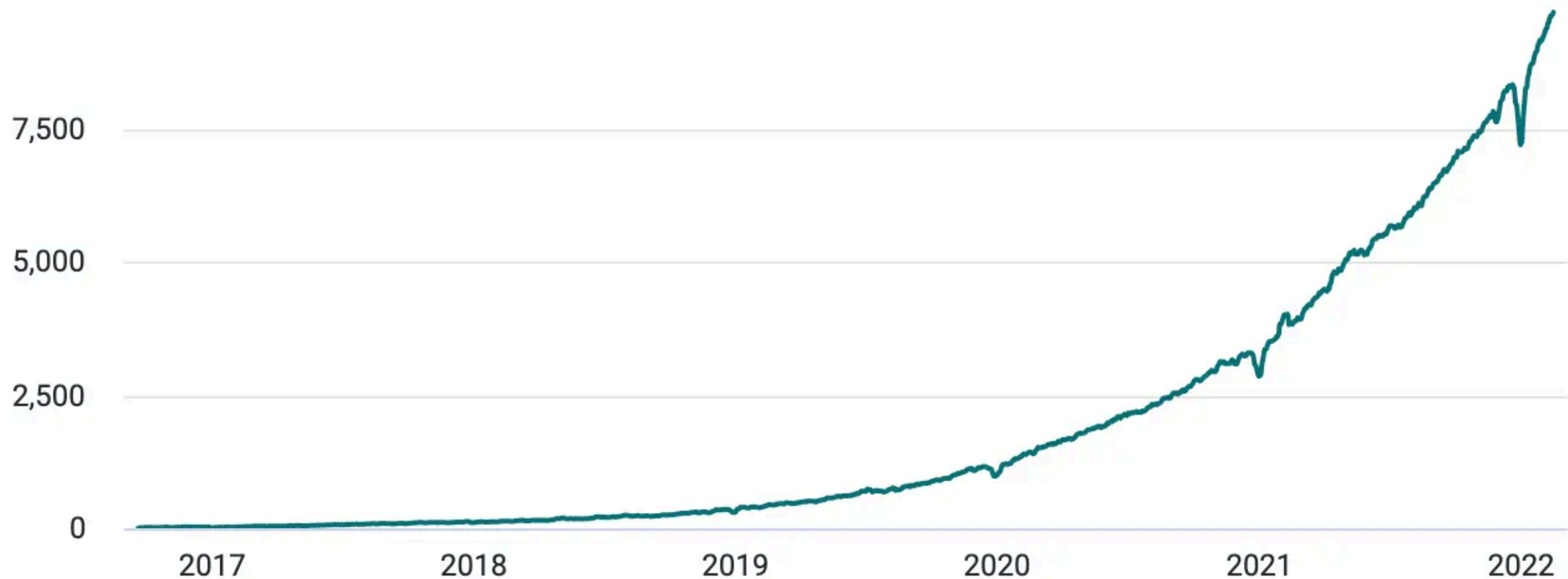


Source: ETL vs. ELT (Wang, 2022)

The modern data stack as we know it (circa 2016)

- **Ingestion:** Fivetran, Stitch
- **Storage:** Bigquery, Databricks, Redshift, Snowflake
- **Transformation:** dbt
- **BI:** Looker, Mode, Periscope, Chartio, Metabase, Redash

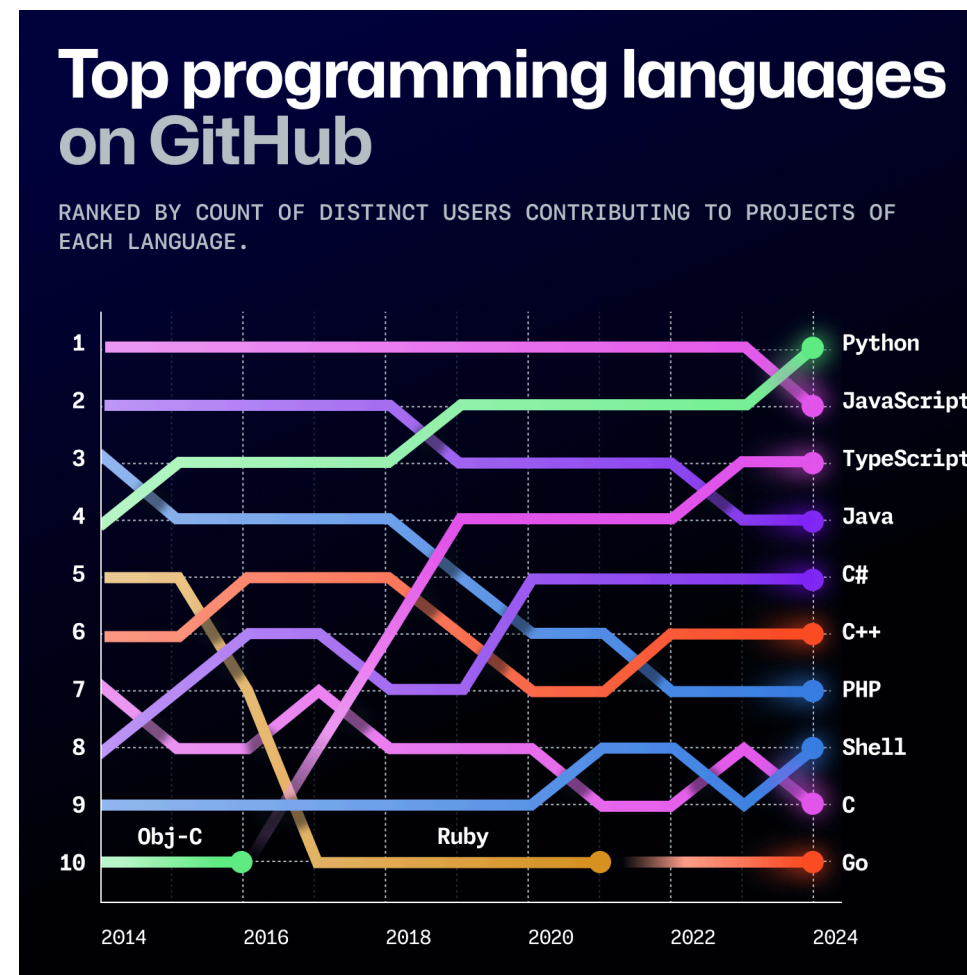
Weekly Active dbt Projects



Source: The next layer of the modern data stack (Handy, 2024)

What about Python?

- Python now rivals SQL in developer popularity
- Second-class support in the modern data stack
 - Still seen as the fallback option for complex cases
- Python-first data pipelines couldn't scale*



Source: Octoverse (GitHub Staff, 2024)

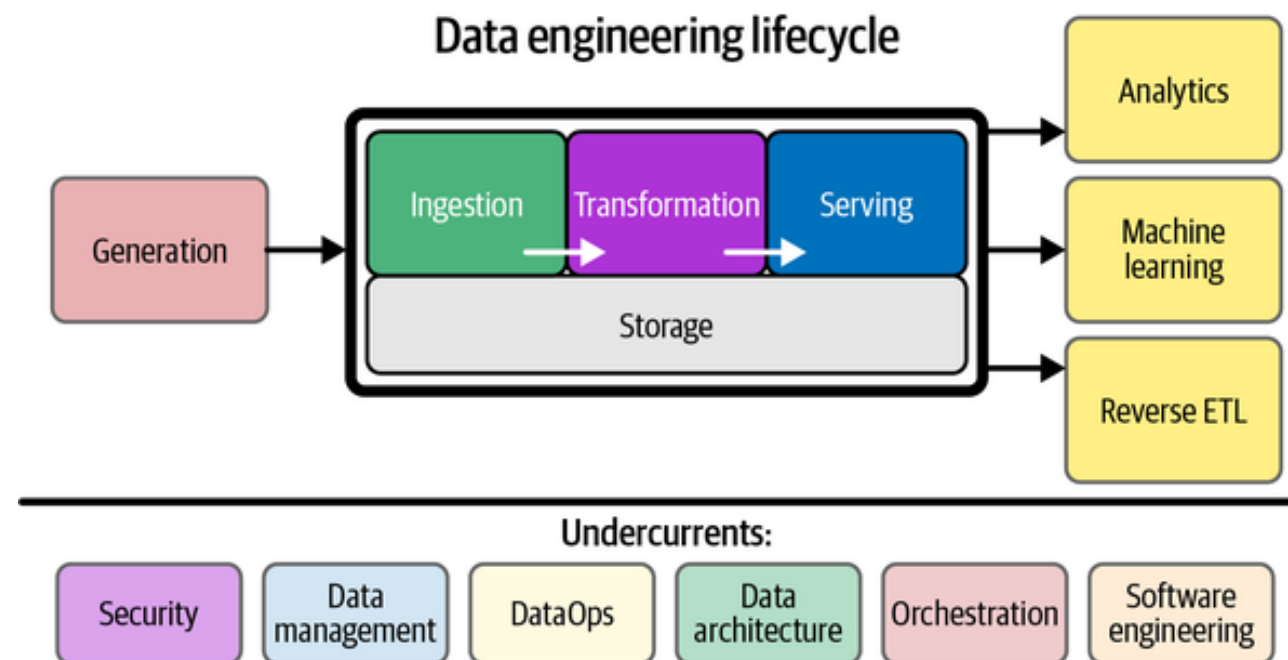
Enter the bird

- **Ibis**: a lightweight Python library for data wrangling
- Provides:
 - Pythonic dataframe API
 - Interfaces to 20+ query languages
 - Deferred execution model
- Enables language interoperability

```
t.filter(t.a > 1).select("a", "b", "c") ≡ SELECT a, b, c FROM t WHERE a > 1
```

Fundamental components of any data stack

- Data storage
- Data ingestion
- Data transformation
- BI and analytics
- Data observability



Source: Fundamentals of Data Engineering (Reis and Housley, 2022)

Data storage

- **Data warehouses:** Ibis supports most common data warehouse providers (e.g. Snowflake, Databricks, BigQuery) as backends
- **Data lakes:** Ibis supports Parquet, CSV, JSON, and more depending on the backend
- **Data lakehouses:** Ibis supports Delta; Iceberg supported by some backends

Data ingestion

- **dlt**: open-source, Python-native EL (and Reverse ETL)
- Features:
 - Loads data from [REST APIs](#), [SQL databases](#), [cloud storage](#), [Python data structures](#), and [more](#)
 - Supports a variety of [popular destinations](#)
 - Also supports [Reverse ETL](#)
 - Built-in [Ibis integration](#)
- Proven part of the modern data stack

dlt example

```
import dlt
from dlt.sources.filesystem import filesystem, read_csv

files = filesystem(bucket_url="gs://filesystem-tutorial", file_glob="encounters*.csv")
reader = (files | read_csv()).with_name("encounters")
pipeline = dlt.pipeline(
    pipeline_name="hospital_data_pipeline",
    dataset_name="hospital_data",
    destination="duckdb",
)

info = pipeline.run(reader)
print(info)
```

Data transformation

- **Kedro**: an open-source Python framework for building reproducible, maintainable, modular data pipelines
- Features:
 - Nodes are vanilla Python functions
 - Native support for reading and writing Ibis tables using Kedro-Datasets
 - Extensible by design
 - Graduate-stage project of the LF AI & Data Foundation

"When I learned about Kedro (while at dbt Labs), I commented that it was like dbt if it were created by Python data scientists instead of SQL data analysts (including both being created out of consulting companies)."

— Cody Peterson, Product Manager @ Ascend.io

From SQL (dbt) to Python (Kedro + Ibis)

SQL (using Jinja):

```
{% set payment_methods = ['credit_card', 'coupon', 'bank_transfer', 'gift_card'] %}  
  
...  
  
{% for payment_method in payment_methods %}  
sum(...) as {{ payment_method }}_amount,  
{% endfor %}
```

Python (using Ibis):

```
payment_methods = ["credit_card", "coupon", "bank_transfer", "gift_card"]  
  
...  
  
total_amount_by_payment_method = {}  
for payment_method in payment_methods:  
    total_amount_by_payment_method[f"{payment_method}_amount"] = ...
```

Data validation

- **Pandera**: the open-source framework for validating Python dataframes
- Supports:
 - pandas, Dask, Modin, PySpark, and now... Ibis!
 - Built-in checks for common validation tasks
 - Custom checks in Ibis syntax
 - Integrations with other tools in the Python ecosystem

Pandera example

```
def total_amount_positive(data: IbisData) -> ibis.Table:
    w = ibis.window(group_by="order_id")
    with_total_amount = data.table.mutate(total_amount=data.table.amount.sum().over(w))
    return with_total_amount.order_by("order_id").select(_.total_amount >= 0)

schema = pa.DataFrameSchema(
    columns={
        "order_id": pa.Column(int),
        "amount": pa.Column(float),
        "status": pa.Column(
            str,
            pa.Check.isin(
                ["placed", "shipped", "completed", "returned", "return_pending"]
            ),
        ),
    },
    checks=[pa.Check(total_amount_positive)],
)
```

BI and analytics

- [Ibis supports the Streamlit `connection\(\)` interface](#)
- It can also be used with most other data visualization libraries
 - pandas/Polars interoperability where needed
- Great fit for teams doing analytics and modeling

Current gaps and limitations

- Still new: some missing features (e.g. upserts in Ibis)
- Community integrations maturing (Kedro-Pandera, orchestrators, OpenLineage)
- Awareness

Why this stack works

- Built organically to solve real user problems
- Unified Python tooling = lower context switching
- Solid fundamentals: based on composable, open standards
- Python is no longer the fallback—it's the foundation

Where to learn more

Let's build better data pipelines—in Python.

- Tools we talked about
 - Ibis: <https://ibis-project.org/>
 - Kedro: <https://kedro.org/>
 - Pandera + Ibis: <https://pandera.readthedocs.io/en/latest/ibis.html>
 - dlt: <https://dlthub.com/docs/intro>
- Putting it all together
 - Jaffle Shop demo: <https://github.com/deepyaman/jaffle-shop/tree/jaffle-stack>
 - Kedro + Ibis: <https://kedro.org/blog/building-scalable-data-pipelines-with-kedro-and-ibis>
 - Kedro + Ibis tutorial: https://www.youtube.com/watch?v=ffDHdtz_vKc
- Dive deeper
 - Does Ibis understand SQL? <https://ibis-project.org/posts/does-ibis-understand-sql/>